

CASE REPORT

Volume 11 Issue 3

*Use of naltrexone for treating binge eating and food obsession related to trauma: a case report and literature review*Rebecca Katlyn Hicks BA¹, Iram Kingson BA, MS¹,
Jordan Gaal DO¹, Suzanne Holroyd MD¹**ABSTRACT**

Impulse-control disorders, including those related to eating behaviors, are complex conditions with limited pharmacologic treatment options. While naltrexone is primarily used for opioid and alcohol dependence, emerging research suggests it may also be effective in managing a variety of impulse-driven conditions. As a pure opioid antagonist, naltrexone's mechanism of action involves the inhibition of opioid receptors on dopaminergic neurons involved in the reward pathway of the brain, thereby reducing the reinforcing effects of substances such as alcohol and opioids. Research suggests that through this mechanism of action, naltrexone shows promise in treating impulsive behaviors also driven by the reward pathway, such as binge eating, purging, gambling, kleptomania, and self-injurious behavior, as well as autism spectrum disorder (ASD) and intellectual disability. This case presents a 28-year-old female with a history of intellectual disability, ASD, schizoaffective disorder, bipolar disorder, and posttraumatic stress disorder (PTSD) who displayed binge eating, hoarding, and aggression related to food following years of food restriction and trauma. After the failure of other medications, naltrexone was prescribed and showed a marked reduction in these behaviors. This case highlights naltrexone's efficacy in treating trauma-associated binge eating and its broader potential in managing impulsivity-related disorders. Ultimately, further research is still needed to determine the proper dosages and duration of treatment and what conditions best indicate the use of naltrexone instead of alternative forms of treatment.

Author affiliations are listed at the end of this article.

Corresponding Author:
Rebecca Katlyn Hicks
Marshall University
Joan C. Edwards
School of Medicine
hicks280@marshall.edu

KEYWORDS

case report, naltrexone, food addiction (FA), impulse control disorder, autism spectrum disorder (ASD), compulsive eating behavior, binge eating, food obsession, trauma, impulse control, intellectual disability, schizoaffective disorder, posttraumatic stress disorder (PTSD), opioid antagonists, compulsive behaviors, impulsivity, trauma-related eating disorders, pharmacotherapy, impulse-driven conditions, reward pathway, psychiatric treatment, eating disorders, obsessive behaviors, food hoarding

INTRODUCTION

Naltrexone is an opioid receptor antagonist primarily used in the treatment of opioid and alcohol dependence. However, emerging research suggests its possible utility in managing a broader range of impulse-driven conditions. As Merrer et al. describe, "opioid receptors are recruited in response to natural rewarding stimuli and drugs of abuse, and both endogenous opioids and their receptors are modified as addiction develops."¹ This understanding has

extended to food-related behaviors: food addiction (FA) has been compared to substance use disorders, suggesting that blocking the opioid reward system in the brain may help alleviate symptoms in certain eating disorders (EDs) and binge eating disorder in particular.²

In both obesity and EDs, food cravings represent a form of anticipatory reward that is regulated by endogenous opioid and mesolimbic dopaminergic systems.³ Current treatment recommendations for



binge eating and obsessive food-related behaviors include psychotherapy —particularly cognitive behavioral therapy (CBT) —and pharmacologic agents such as selective serotonin reuptake inhibitors (SSRIs) or lisdexamfetamine.⁴ While these approaches aim to reduce the frequency of binge eating episodes, lisdexamfetamine is currently only approved for use in adults with moderate to severe binge eating disorders.⁴ This treatment paradigm aligns with research suggesting that opioid antagonists may help reduce reward sensitivity associated with hyperpalatable food intake.⁴ Given this mechanism, pharmacologic strategies that target reward pathways have gained interest in managing obesity and related disordered eating behaviors. Notably, the weight-reducing effects of naltrexone appear to be more pronounced when combined with bupropion than when either medication is used alone, supporting its potential utility in cases of obesity driven by reward-based eating patterns.⁵

These findings support the broader therapeutic role of naltrexone in conditions characterized by dysfunctional reward processing and compulsive behaviors. Building on this, recent studies have explored its use in a range of impulse control disorders beyond eating-related pathology. Although symptoms of impulse control disorders are generally refractory to psychotherapeutic or pharmacologic treatments, recent studies suggest that opioid antagonists may reduce human urges, one of the core symptoms of impulse control disorders.⁶ In one study, 15 patients who had impulse control disorder were prescribed naltrexone for up to 9 months.⁶ It was generally well tolerated and had no hepatic side effects.⁶ Naltrexone appears to reduce urge-related symptoms and decrease behaviors such as pathological gambling, and the effect appears to be sustained.⁶ Notably, a standard dose of 50 mg/day was not adequate for most patients; higher doses were required to achieve clinical benefit.⁶

Earlier studies show that naltrexone can also be effective in decreasing hyperactivity and impulsive behaviors in ASD.⁷ Additionally, symptoms like self-injury behaviors, hyperactivity, and stereotyped and ritualistic conduct appeared to be improved in

a subgroup of children with the opiate antagonist.⁷ However, new controlled studies still need to be done before recommending naltrexone for patients with autism.

In an 8-week, double-blind, placebo-controlled trial to evaluate the safety and efficacy of oral naltrexone for kleptomania, 25 individuals with diagnosed kleptomania were randomized to naltrexone (ranging from 50mg to 150mg per day) or placebo; subjects assigned to naltrexone had significantly greater reductions in stealing urges and stealing behavior.⁸ In a study on the effects of opioid antagonists on self-injury in adults with mental retardation, during naltrexone administration, there were fewer days with frequent head-banging and self-biting, less frequent blows to the head, and less frequent self-biting.⁹

The case in this report is unique due to the use of naltrexone in managing binge-eating behavior in a patient with ASD, a population for whom pharmacologic treatment of impulse-control-related EDs is not well studied. This report adds to the growing body of literature exploring naltrexone's utility beyond substance use, particularly in addressing food addiction-like symptoms.

CASE PRESENTATION

This case presents a 28-year-old female with a history of intellectual disability, ASD, schizoaffective disorder, bipolar type, and PTSD. Per records, she demonstrated psychosis by at least age 16. The patient's parents reportedly did not believe in her diagnosis and refused conventional treatments, believing that she had instead been "possessed by the devil." On numerous occasions for several years, the patient's mother would physically restrain the patient and deprive her of food and water. The patient was eventually admitted to an inpatient facility after running away from home and being found in a psychotic state.

After 8 years in the inpatient setting, it was reported that the patient exhibited aggressive and threatening behaviors towards herself, staff, and other patients. She was also reported to have food-



hoarding behaviors, hyperfixation on appetite, food gorging, and psychogenic polydipsia. Also noted were episodes of stealing food from other patients and aggression at mealtimes. She described intense urges to eat large quantities of food, often accompanied by a sense of loss of control and subsequent distress, multiple times per week. These symptoms were self-reported and align with DSM-5 criteria for binge-eating disorder. The patient had previously seen a nutritionist and therapist, but these interventions had yielded limited improvement in her symptoms. She also participated in online support groups, though engagement was inconsistent due to anxiety symptoms and her sensory sensitivity.

Initially, the patient was prescribed medications for impulse control, aiming to lessen food behavior. Vyvanse (lisdexamfetamine) was unavailable in the pharmacy, and the patient did not tolerate Ritalin (methylphenidate). The prominence of food-related aggression and obsessive behaviors, along with the failure of other central nervous system (CNS) stimulant medications, led to the decision to trial the patient on 50 mg of naltrexone daily to target a different pathway of impulse control.

The patient was started on naltrexone at a dose of 50 mg daily and reported decreased urges related to FA. After 60 days of treatment, the patient had no reported episodes of food aggression, hyper-fixation on appetite, or food theft. With the addition of naltrexone to the patient's therapeutic strategies, the patient has been able to make strides in impulse control and become closer to getting discharged from the inpatient setting back into the community.

DISCUSSION

In this patient's case, prolonged deprivation of food and water manifested as food-obsessive impulsive behaviors observed in the inpatient psychiatric setting. She exhibited a hyperfixation on appetite characterized by food hoarding and gorging, plus psychogenic polydipsia, which is commonly seen in patients with chronic schizophrenia.¹⁰

Given the role of the opioid reward system in compulsive binge eating and purging, naltrexone—an opioid antagonist—has been shown to reduce these behaviors by attenuating the reinforcing effects of food.¹¹ Multiple studies support naltrexone's efficacy in decreasing appetite and binge eating, likely due to its ability to block the anticipatory reward associated with food cravings and consumption via the opioid pathway.¹¹

Food restriction is known to enhance the rewarding effects of both food and drugs, which may explain how the opioid system contributes to both adaptive and pathological behaviors. For instance, a study of trauma-exposed male and female veterans found that PTSD was significantly associated with eating disorder symptoms and food addiction (FA), highlighting the link between trauma and reward-driven eating behaviors.¹²

The overlap between FA and substance use disorders suggests that similar treatment approaches may be beneficial. While further investigation is needed, evidence-based therapies used in addiction—such as CBT, dialectical behavior therapy (DBT), motivational interviewing (MI), motivational enhancement therapy (MET), mindfulness-based therapies (MBT), family therapies, and pharmacotherapies—may serve as beneficial interventions in treating FA.¹²

Although naltrexone has demonstrated efficacy in treating impulse-control behaviors, this case highlights explicitly its potential utility in managing compulsive eating behaviors resembling food addiction (FA). Notably, it is important to distinguish these behaviors from other impulse-control disorders, such as gambling or kleptomania, which were not present in this patient.

The patient's co-occurring diagnosis of autism spectrum disorder (ASD) may have played a role in her restricted interests, sensory sensitivities, and difficulties with emotional regulation, all of which could have contributed to her maladaptive eating behaviors and influenced her response to treatment. Routine-oriented thinking patterns commonly seen in ASD may also have affected the sustainability of behavioral changes during intervention. Strengths of this case report include the novel



documentation of naltrexone use in a patient with ASD, an underrepresented population in existing literature, and the inclusion of patient-reported outcomes, which provide valuable insight into subjective treatment response. Limitations include the absence of standardized clinical assessments to quantify binge-eating severity and treatment response, and the potential influence of nonspecific treatment effects such as placebo response or environmental changes within the inpatient setting.

FUTURE DIRECTIONS

Although several studies have shown promising results for a wide variety of uses for naltrexone, there is much left to research regarding its safety and outcomes. Naltrexone's effects impact the whole body and can mimic the side effects of opioid withdrawal due to the blockade of receptors.¹³ Common side effects include nausea, headaches, tiredness, restlessness, and abdominal cramping.¹³ More serious side effects, such as hepatotoxicity, are more likely with higher doses of naltrexone, which may be required in some patients who do not respond to low doses.¹⁴ In one study, 16 individuals with bulimia consented to a 6-week trial of naltrexone, receiving either standard dosages of 50 to 100 mg each day or high dosages of 200 to 300 mg each day.¹⁵ At the end of 6 weeks, individuals in the low-dose group had no significant change in their binge eating or purging frequency, while individuals in the high-dose group had significant reductions in both behaviors.¹⁵ Four individuals in the low-dose group who crossed over to high-dose naltrexone at the end of the study went on to experience significant reductions in binge eating and purging.¹⁵ These findings support the potential utility of opiate blockade in treating bulimia but suggest that dosages of naltrexone greater than those needed to block exogenous opiates may be required for therapeutic efficacy in reducing binge eating and purging.¹⁵ Further research on proper dosage and duration of treatment is needed.

It appears that results found when using naltrexone in EDs are quite different when considering the duration of the study, the dose of naltrexone given to patients, the age range of the patients, and the kind of ED.¹⁵ There is a discrepancy between the short-

term effect of opiate antagonists on food intake and the lack of effect of long-term naltrexone treatment on body weight in human subjects.¹⁵ As the doses chosen in the different studies on naltrexone in EDs vary, it is possible that higher dosages may act by a different mechanism compared with lower dosages.⁷

It is important to note that in adults, results found with the use of naltrexone in EDs are different when considering the duration and the dosage of the treatment and the kind of ED (bulimia, binge eating, or anorexia nervosa).⁷ Previous studies have demonstrated that naltrexone reduces binge eating and purging in adults, yet evidence for its use in adolescents is lacking; therefore, studies in children and adolescents are needed.⁷

Although this case highlights the utilization and response of low-dose naltrexone in the pharmacological treatment of food bingeing and obsession, further research could be directed toward the efficacy of pharmacological treatment versus psychotherapy to address the underlying trauma leading to food obsession.¹⁵ The lack of evidence on the effectiveness of psychotherapy treatment in addition to psychotropic medications compounds the difficulties in stabilizing individuals with EDs.¹⁶

CONCLUSION

In conclusion, both recent research and the case of this patient suggest the efficacy of naltrexone in treating various forms of impulse-driven conditions, including food addiction. This case also supports its potential use in ASD patients with disordered eating behaviors. While these findings are hopeful, further research is needed to determine the optimal dosage and duration of treatment and whether other avenues of treatment (isolated or adjunct) could be just as, if not more, beneficial in treating binge eating and food obsession related to trauma. While the use of naltrexone has been relatively limited to certain conditions, the growing amount of research supporting its use for various conditions shows great promise in treating conditions with limited treatment options up to this point.



INFORMED CONSENT

The authors confirm that written informed consent was obtained from the patient for the publication of this case report.

CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

ACKNOWLEDGMENTS

The authors would like to thank the Department of Psychiatry at Marshall University Joan C. Edwards School of Medicine for their support in this research. They give special thanks to the medical students who contributed to this work as part of their summer research project. No funding was received for this research.

AUTHOR AFFILIATIONS

1. Marshall University Joan C. Edwards School of Medicine, Huntington, West Virginia

REFERENCES

1. Le Merrer J, Becker JA, Befort K, Kieffer BL. Reward processing by the opioid system in the brain. *Physiol Rev*. Oct 2009;89(4):1379–412. doi:10.1152/physrev.00005.2009
2. Chiara G, Pietra C. The role of the opioid system in binge eating disorder. *CNS Spectr*. Oct 2015;20(6):537–545. doi:10.1017/S1092852915000668
3. Volkow ND, Wang GJ, Tomasi D, Baler RD. The addictive dimensionality of obesity. *Biol Psychiatry*. May 1 2013;73(9):811–8. doi:10.1016/j.biopsych.2012.12.020
4. Brownley KA, Berkman ND, Peat CM, et al. Binge-Eating Disorder in Adults: A Systematic Review and Meta-analysis. *Ann Intern Med*. Sep 20 2016;165(6):409–20. doi:10.7326/M15-2455
5. Tek C. Naltrexone HCl/bupropion HCl for chronic weight management in obese adults: patient selection and perspectives. *Patient Prefer Adherence*. 2016;10:751–9. doi:10.2147/PPA.S84778
6. Kim SW. Opioid antagonists in the treatment of impulse-control disorders. *J Clin Psychiatry*. Apr 1998;59(4):159–64.
7. Chabane N, Leboyer M, Mouren-Simeoni MC. Opiate antagonists in children and adolescents. *Eur Child Adolesc Psychiatry*. 2000;9 Suppl 1:144–50. doi:10.1007/s007870070018
8. Grant JE, Kim SW, Odlaug BL. A double-blind, placebo-controlled study of the opiate antagonist, naltrexone, in the treatment of kleptomania. *Biol Psychiatry*. Apr 1 2009;65(7):600–6. doi:10.1016/j.biopsych.2008.11.022
9. Thompson T, Hackenberg T, Cerutti D, Baker D, Axtell S. Opioid antagonist effects on self-injury in adults with mental retardation: response form and location as determinants of medication effects. *Am J Ment Retard*. Jul 1994;99(1):85–102.
10. Bhatia MS, Goyal A, Saha R, Doval N. Psychogenic Polydipsia - Management Challenges. *Shanghai Arch Psychiatry*. Jun 25 2017;29(3):180–183. doi:10.11919/j.issn.1002-0829.216106
11. Stancil SL, Adelman W, Dietz A, Abdel-Rahman S. Naltrexone Reduces Binge Eating and Purging in Adolescents in an Eating Disorder Program. *J Child Adolesc Psychopharmacol*. Nov 2019;29(9):721–724. doi:10.1089/cap.2019.0056
12. Brewerton TD. Food addiction as a proxy for eating disorder and obesity severity, trauma history, PTSD symptoms, and comorbidity. *Eat Weight Disord*. Jun 2017;22(2):241–247. doi:10.1007/s40519-016-0355-8
13. Kim PS, Fishman MA. Low-Dose Naltrexone for Chronic Pain: Update and Systemic Review. *Curr Pain Headache Rep*. Aug 26 2020;24(10):64. doi:10.1007/s11916-020-00898-0
14. Opioids, Opioid Antagonists. *LiverTox: Clinical and Research Information on Drug-Induced Liver Injury*. 2012.
15. Jonas JM, Gold MS. The use of opiate antagonists in treating bulimia: a study of low-dose versus high-dose naltrexone. *Psychiatry Res*. May 1988;24(2):195–9. doi:10.1016/0165-1781(88)90062-5
16. Valbrun LP, Zvonarev V. The Opioid System and Food Intake: Use of Opiate Antagonists in Treatment of Binge Eating Disorder and Abnormal Eating Behavior. *J Clin Med Res*. Feb 2020;12(2):41–63. doi:10.14740/jocmr4066

